

HCES 2023-24

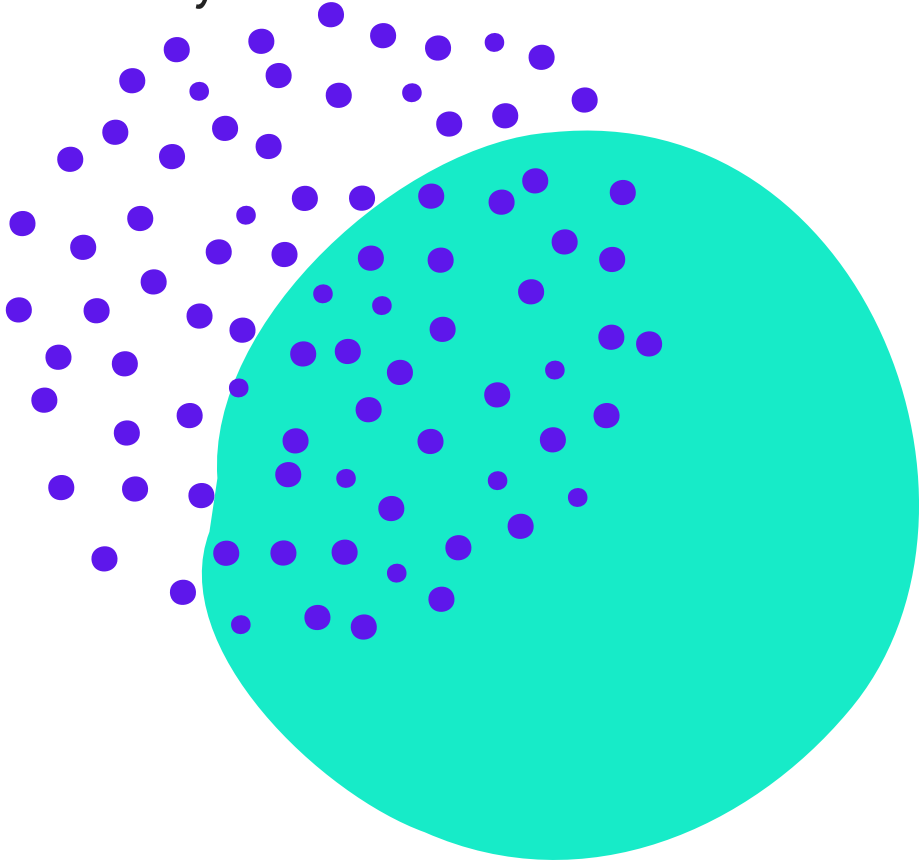
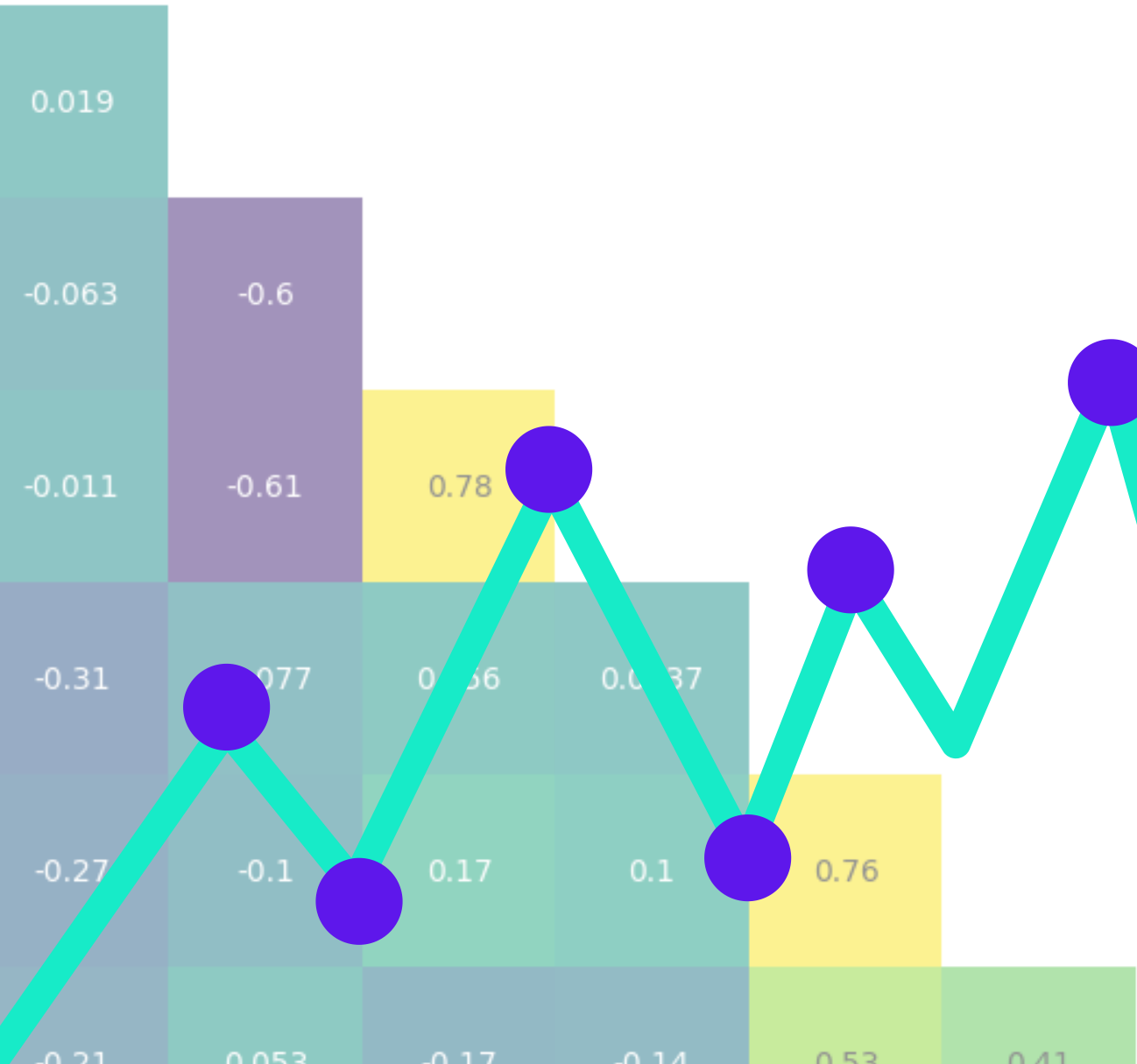
COMPREHENSIVE ANALYSIS

CORRELATION ANALYSIS

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PROCEDURE

Filter Features Heatmap

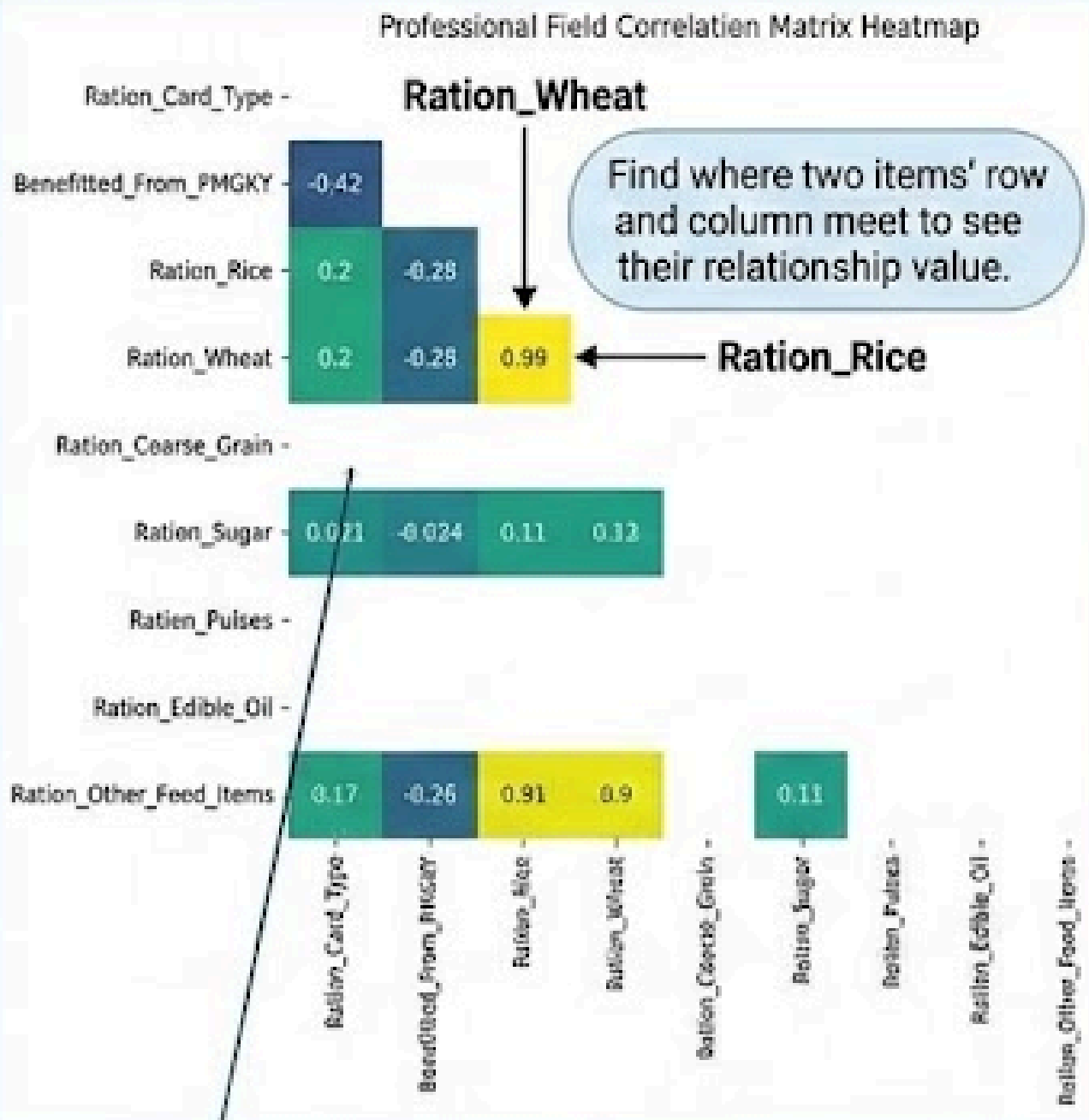
- Based on the features
- From the different levels, cherry-pick the different features acc. to the themes
- Find the correlation
- Plot the heatmap

Level wise Feature Heatmap

- Select only required features
- Find the correlation
- Mask the upper triangular
- Then plot the Heatmap with the value, color bar along with the columns

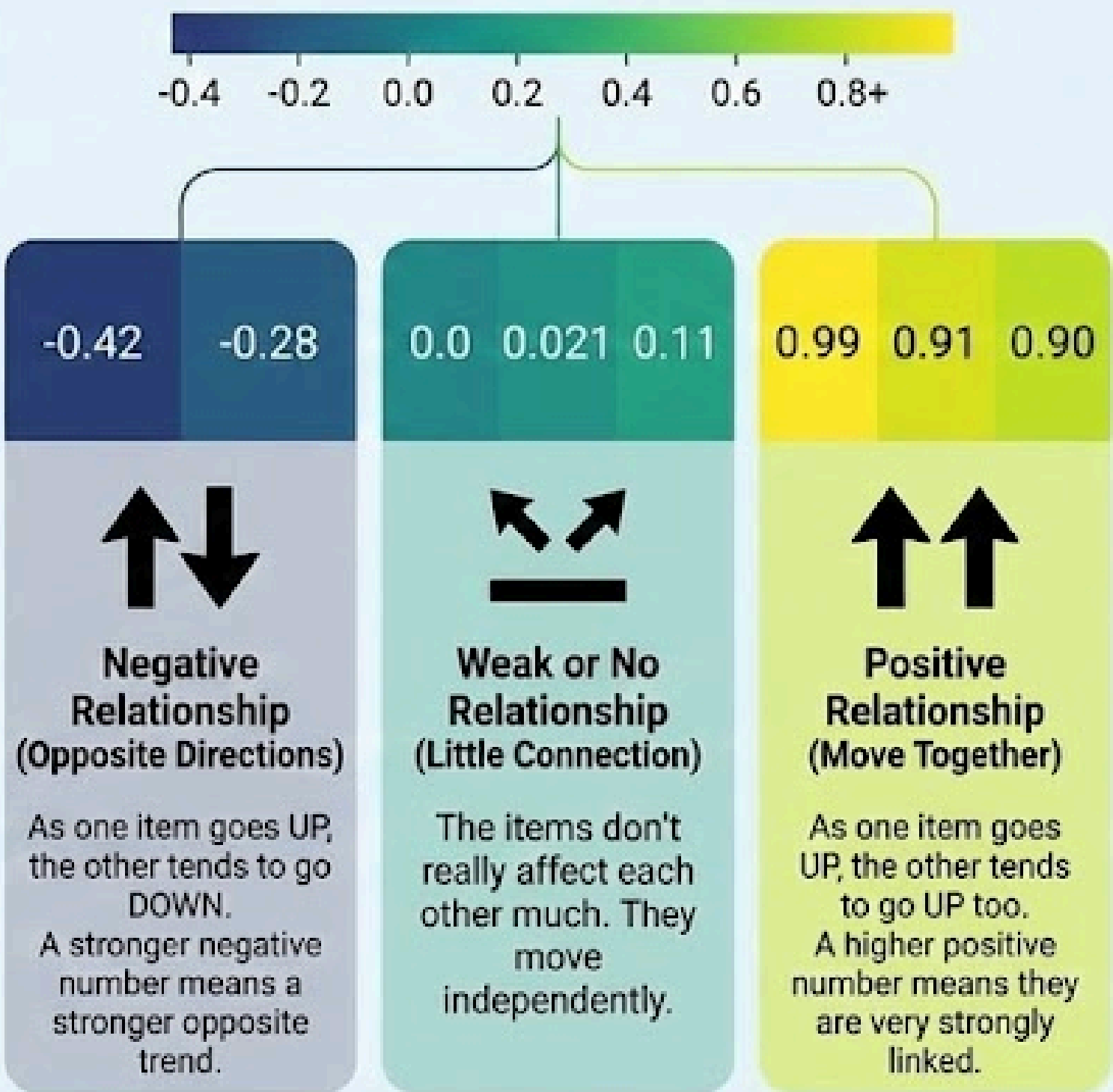
Understanding the correlation Heatmap: A Non-Technical Guide

1. How to Read the Map (The Grid)



2. The Missing Value (Blank Space):
No Data/No Correlation for this pair.

3. The "Corr" Value (Correlation): What it Means

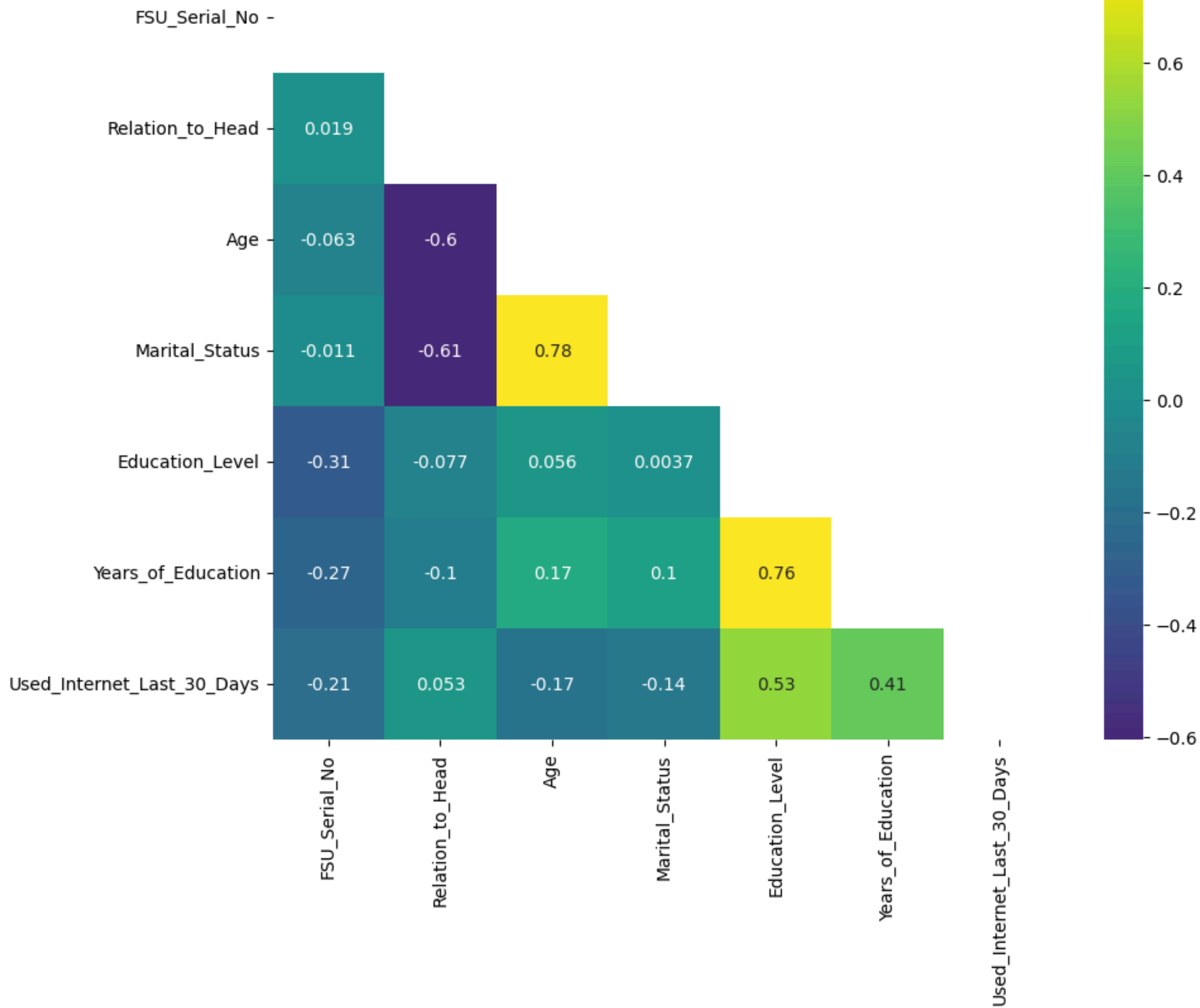


4. Key Takeaways from This Map

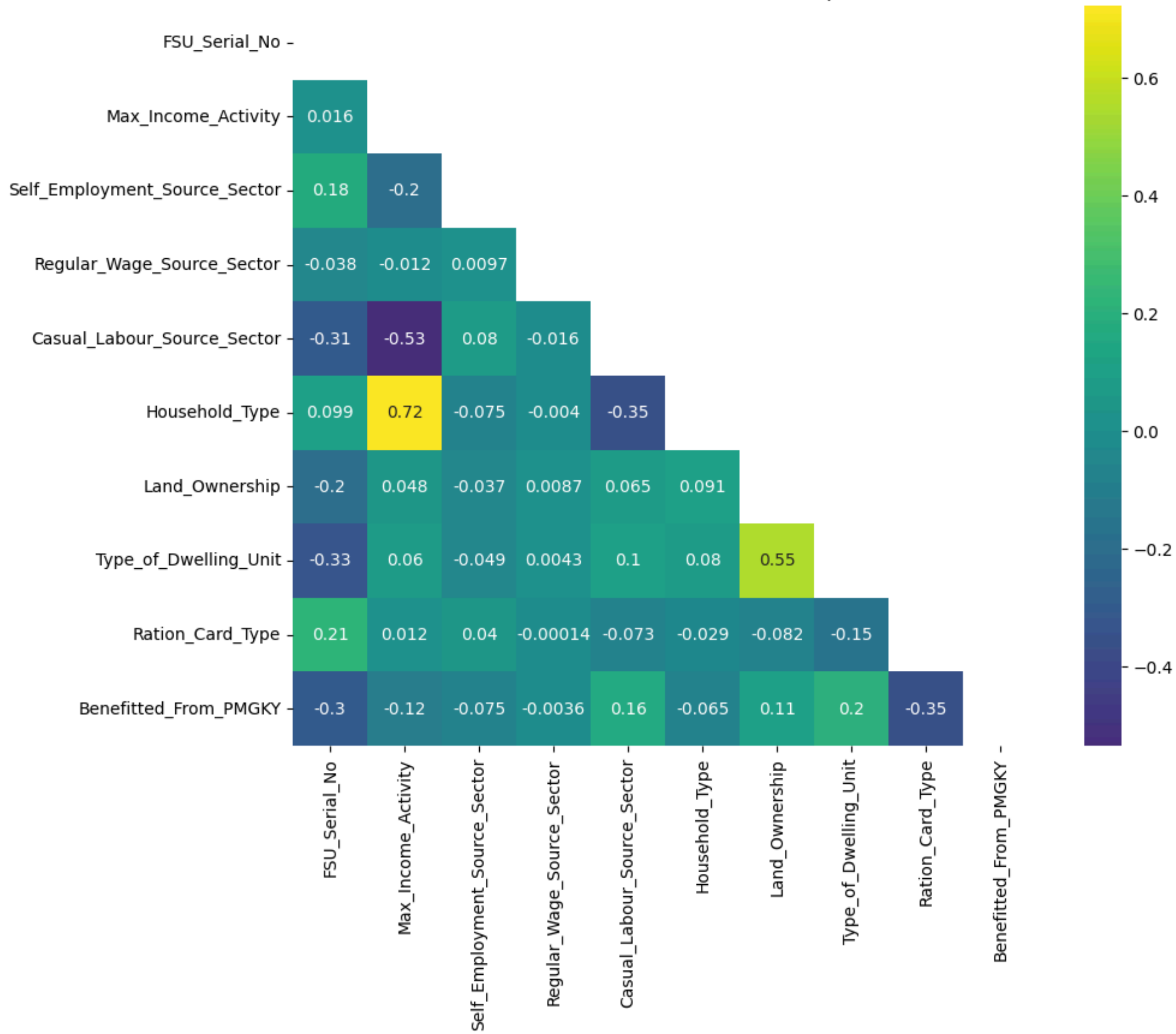
- Strongest Positive Links:** Ration_Wheat & Ration_Rice (0.99), Ration_Other_Food_Items & Ration_Rice (0.91), Ration_Other_Food_Items & Ration_Wheat (0.90). These items are highly connected.
- Notable Negative Link:** Benefitted_From_PMGKY & Ration_Card_Type (-0.42). Having one often means less of the other.
- Missing Data:** Many pairs have no correlation value shown (blank boxes), meaning their relationship is unknown or not calculated here.

LEVEL FEATURES

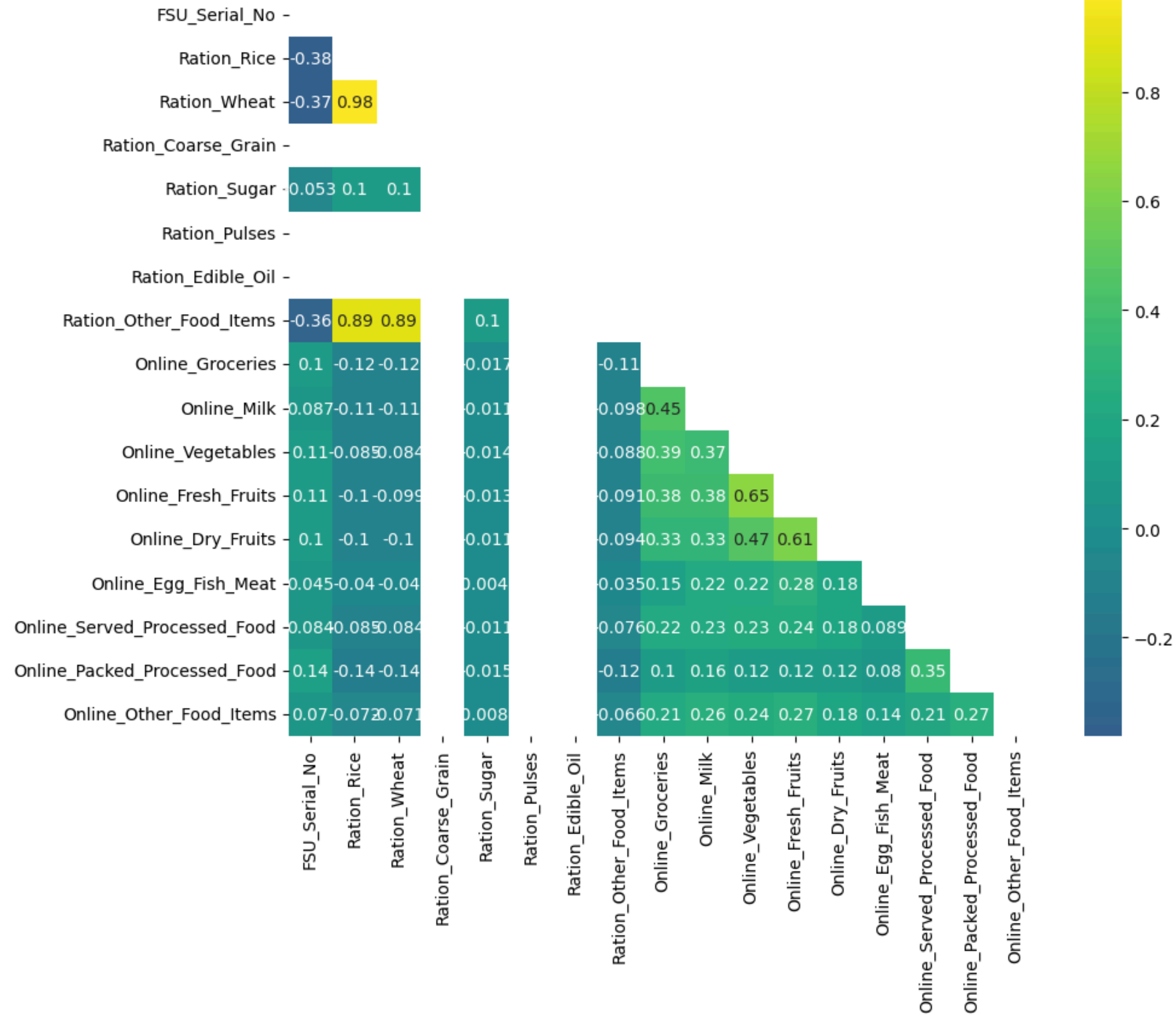
Level 2 Correlation Matrix Heatmap)



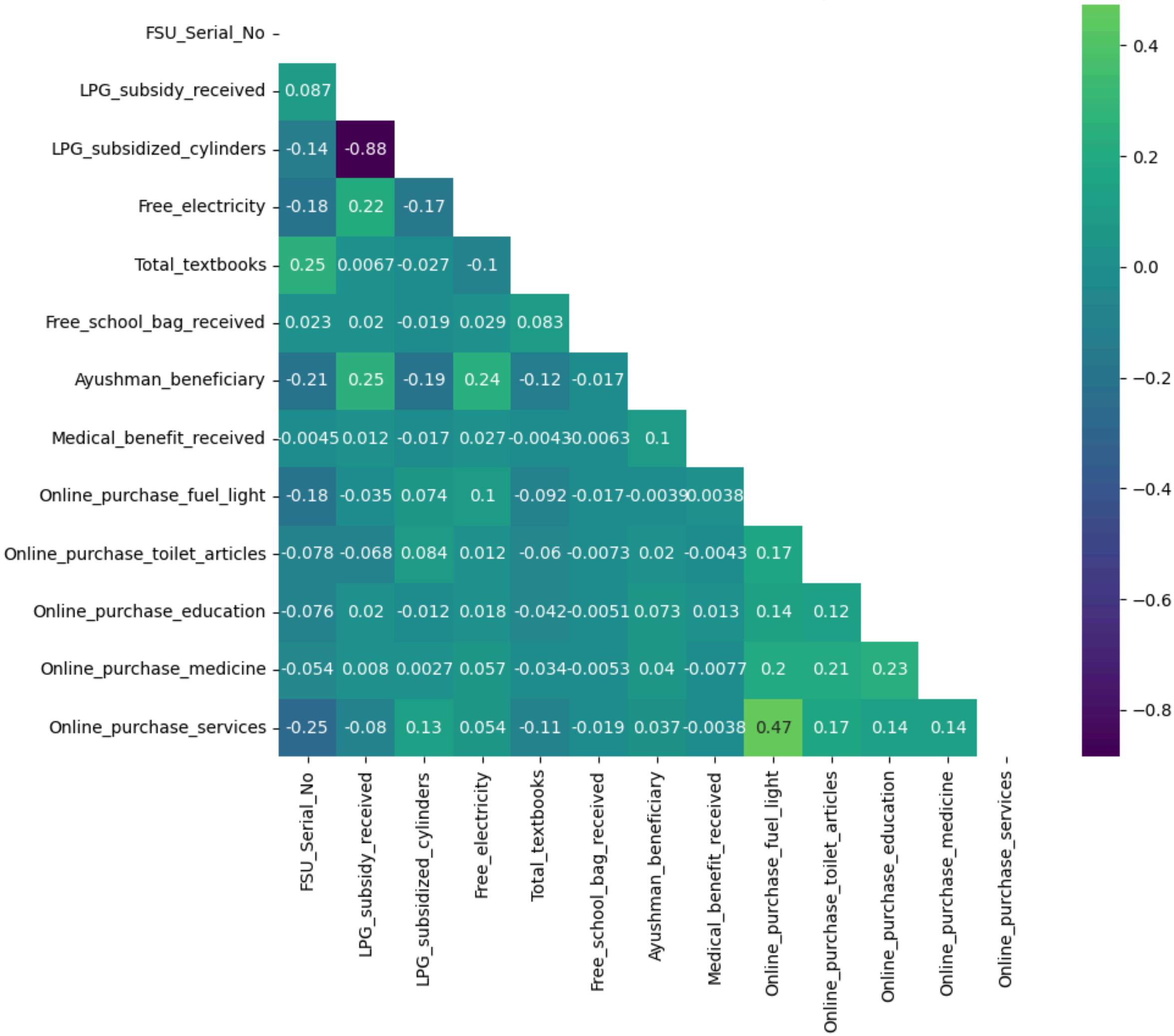
Level 3 Correlation Matrix Heatmap)



Level 04 Correlation Matrix Heatmap)

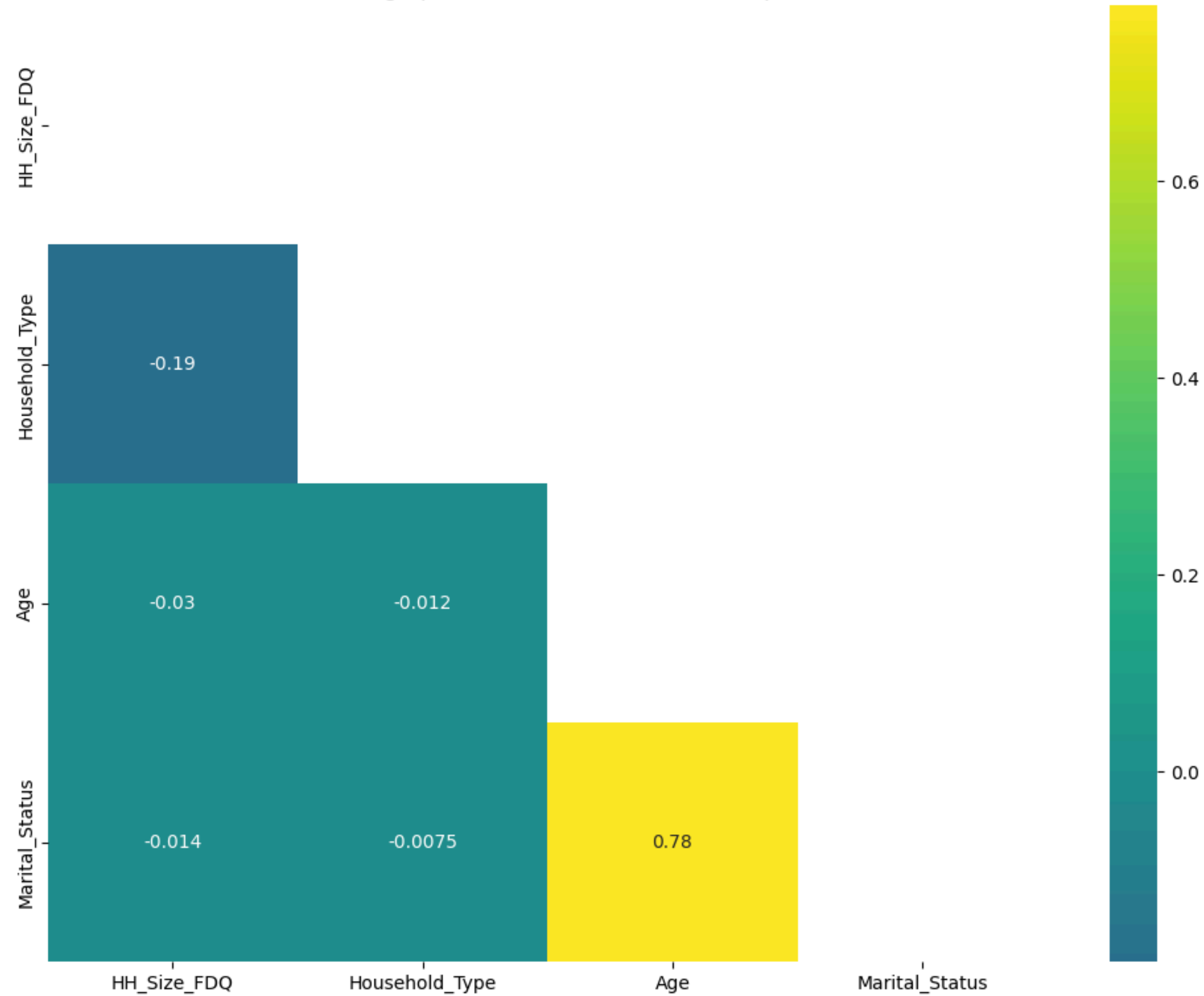


Level 7 Correlation Matrix Heatmap)

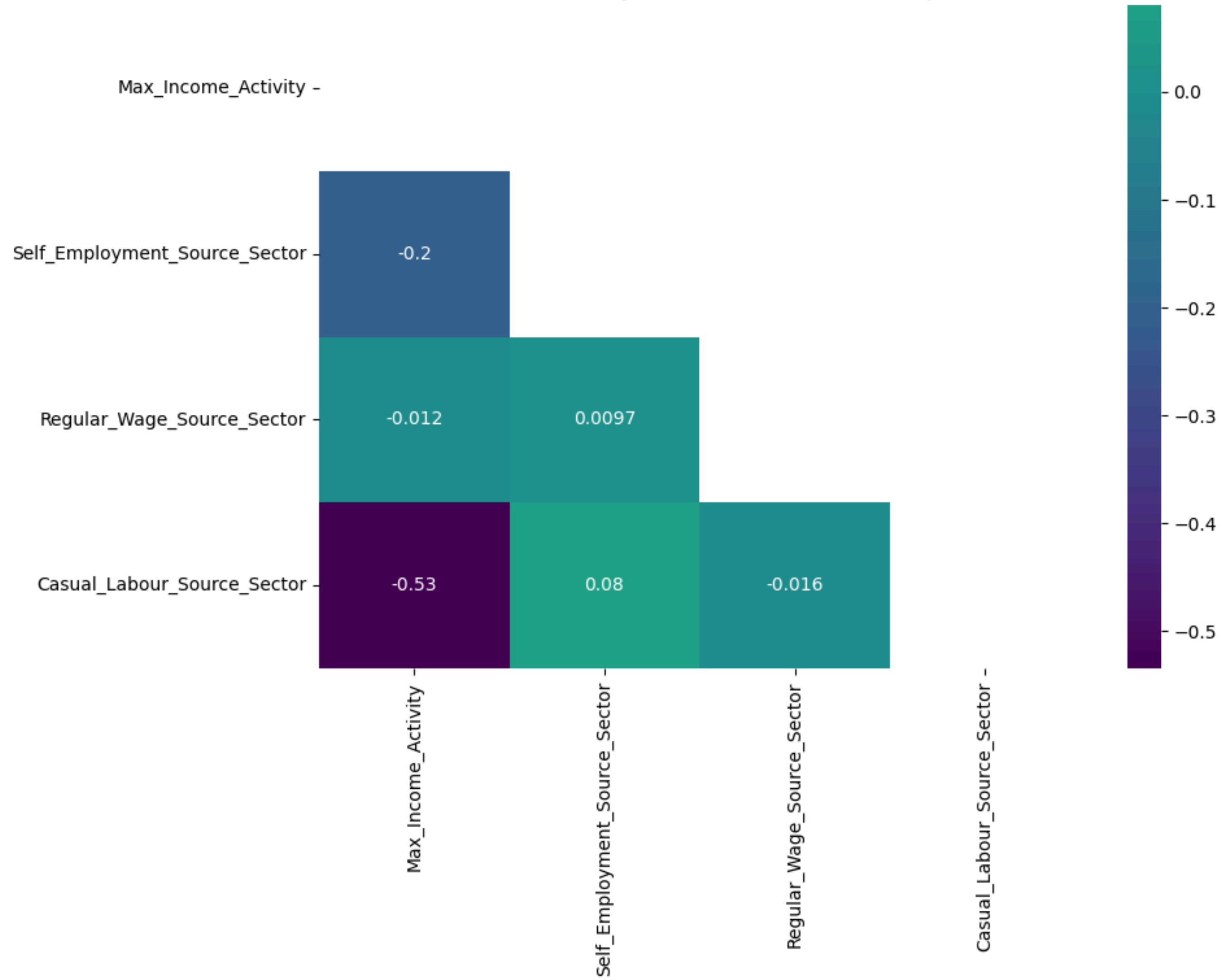


FILTER FEATURES

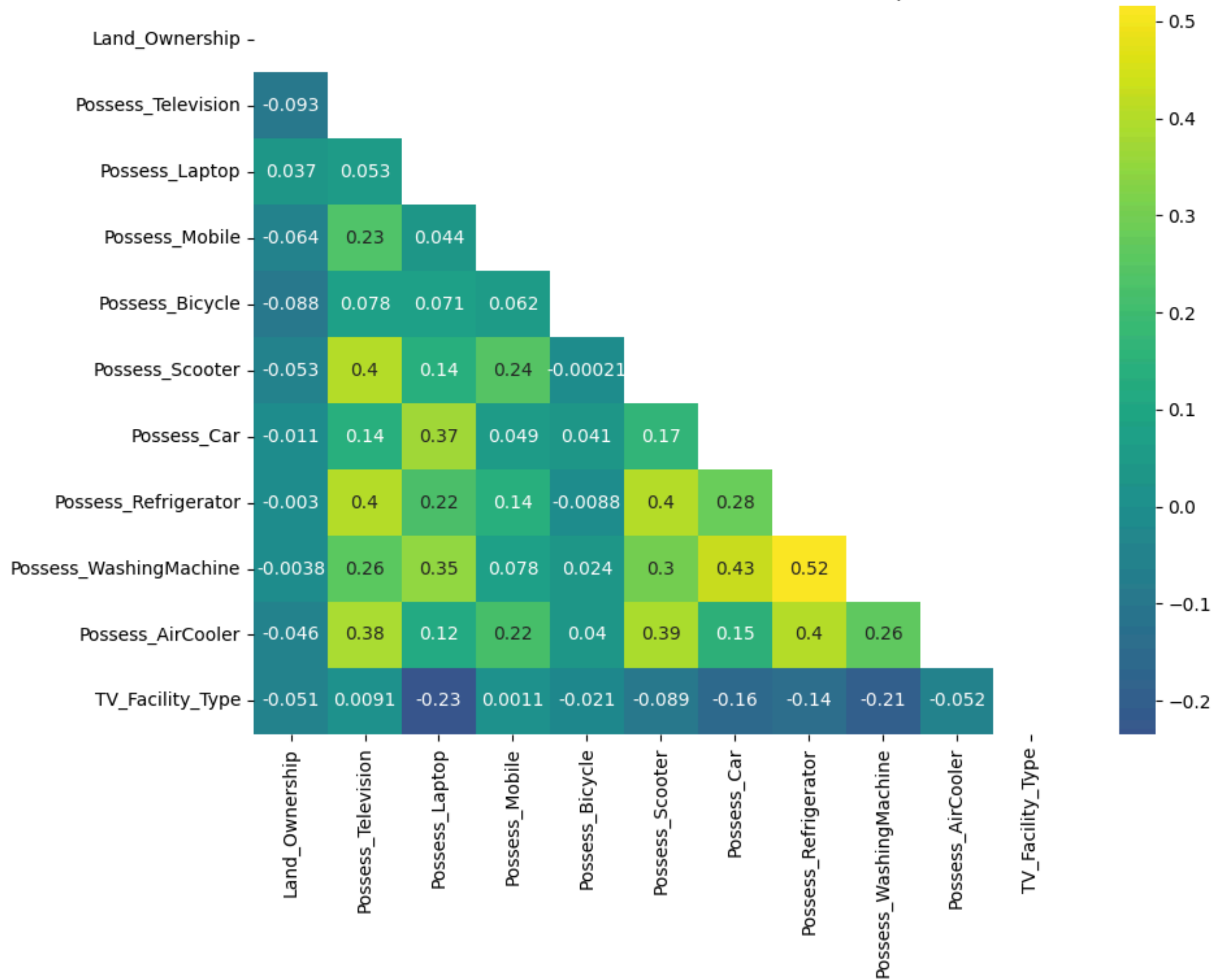
Demographic Correlation Matrix Heatmap

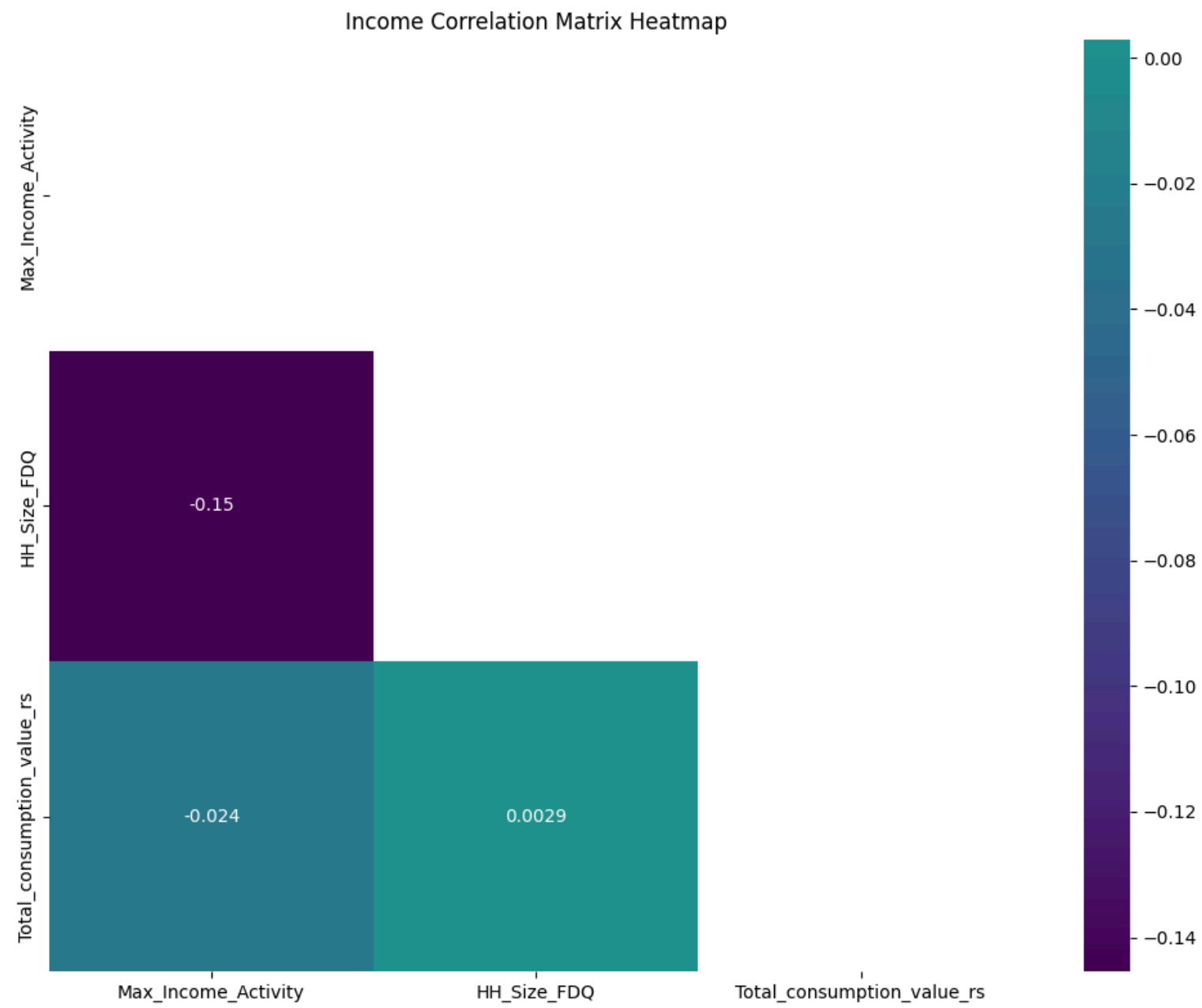


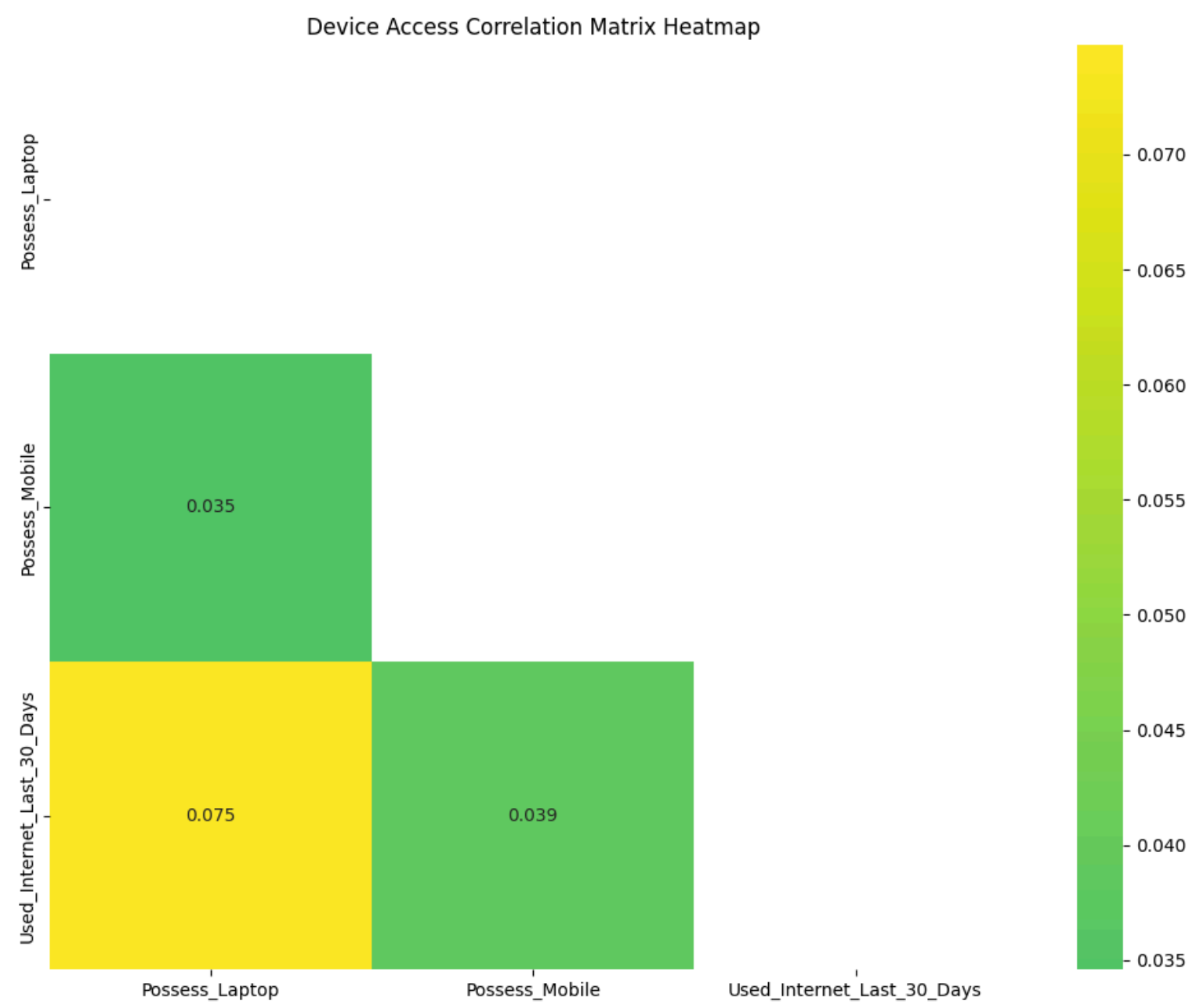
Income Stability Correlation Matrix Heatmap

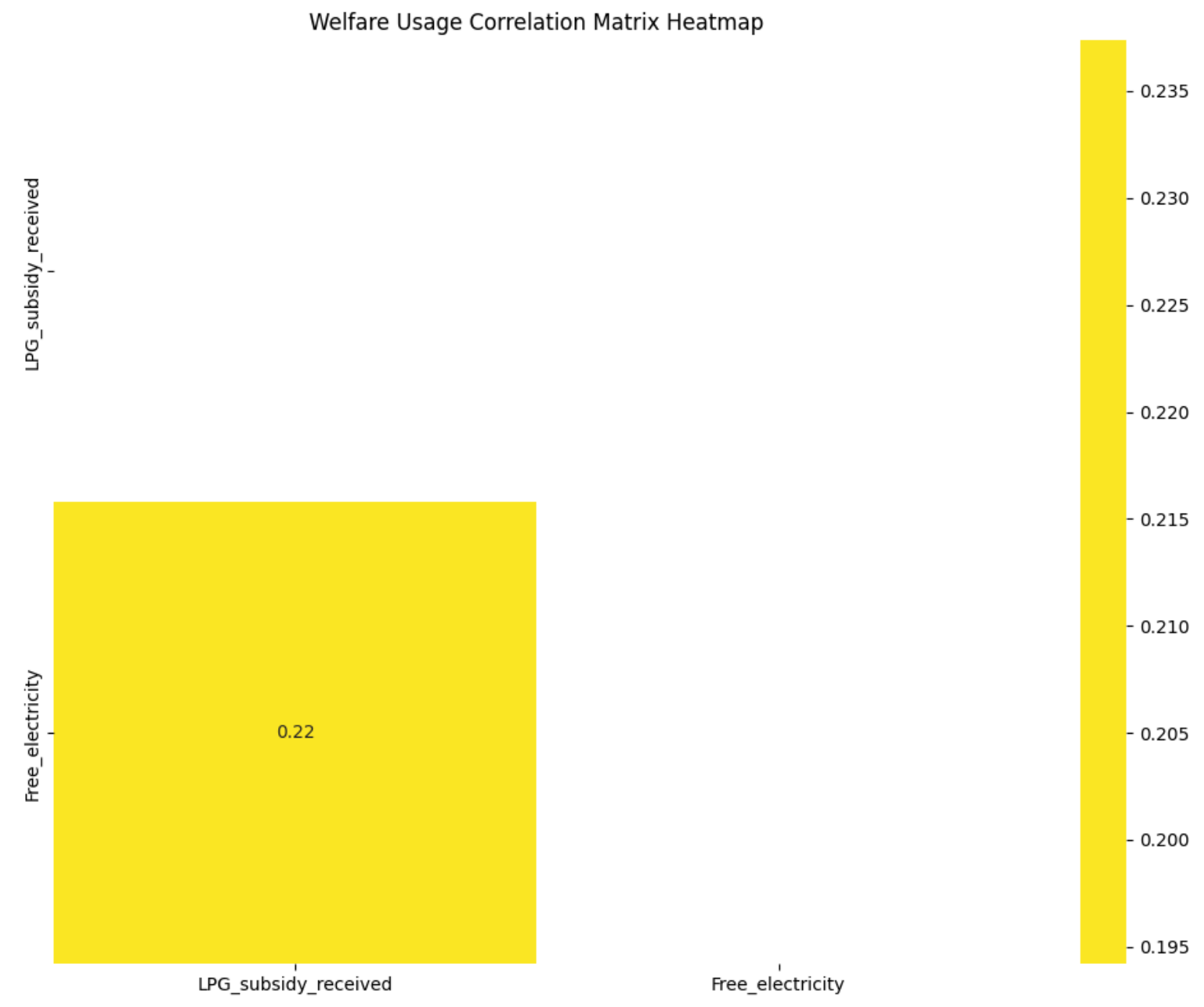


Number of Assets Correlation Matrix Heatmap

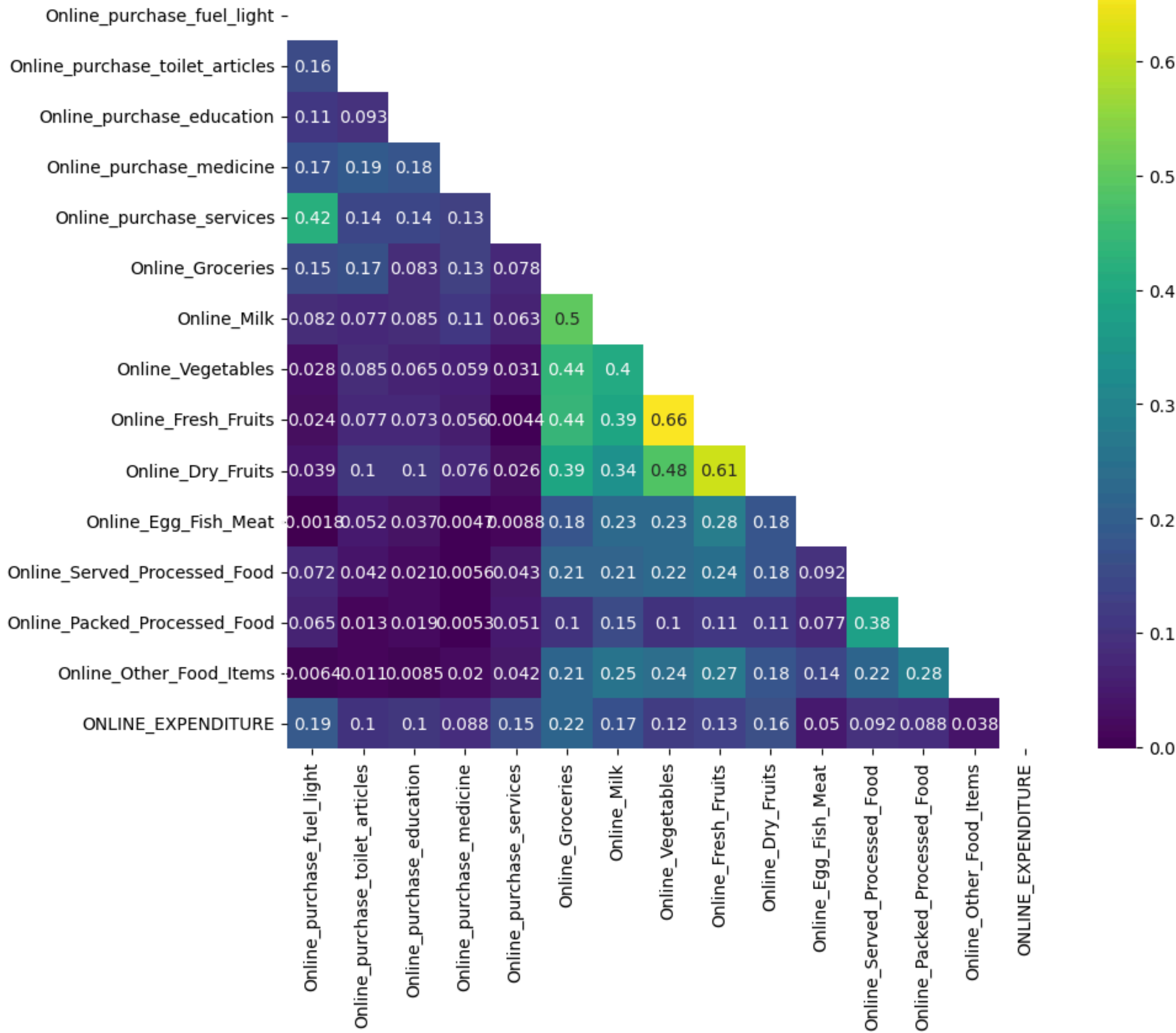




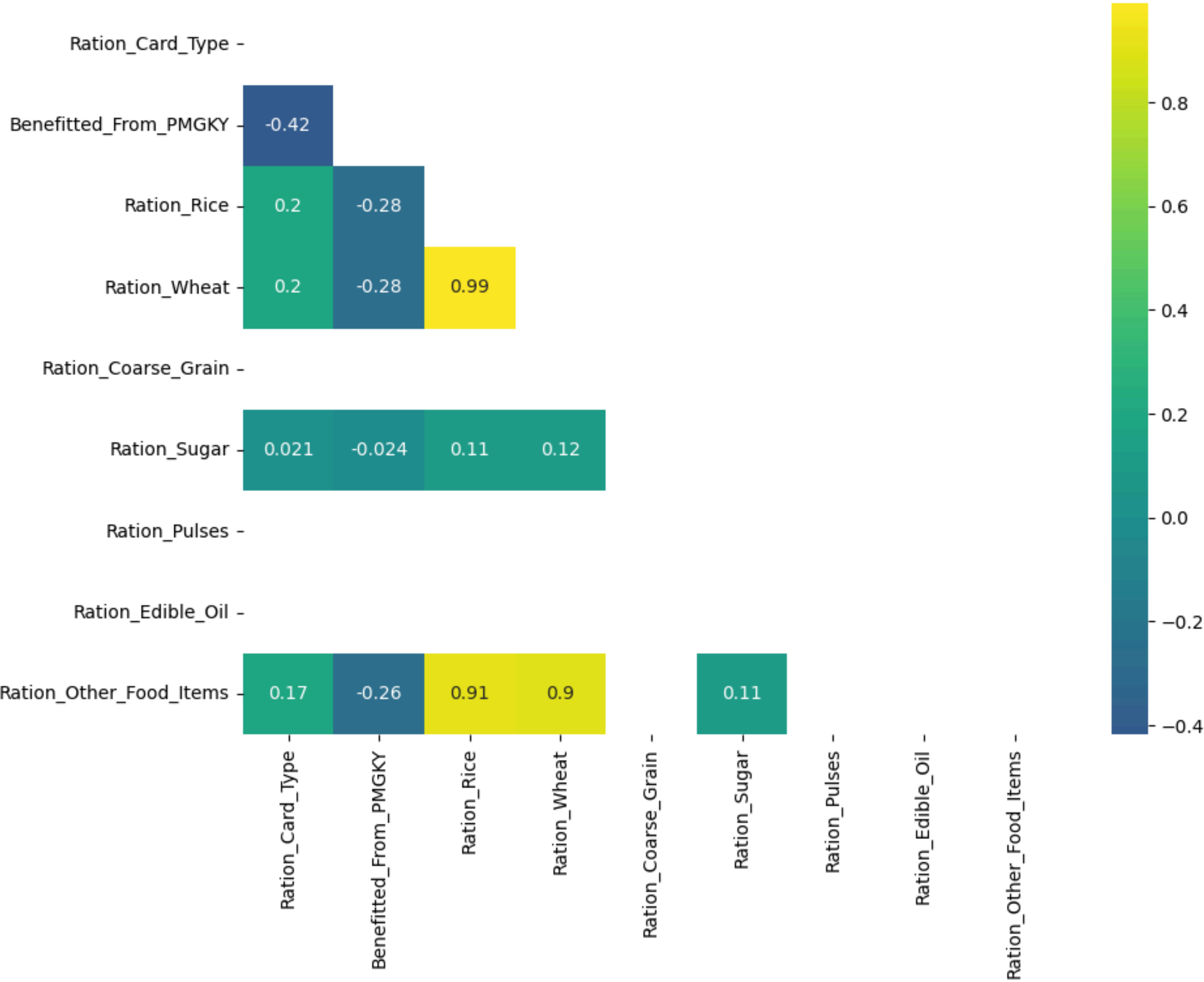




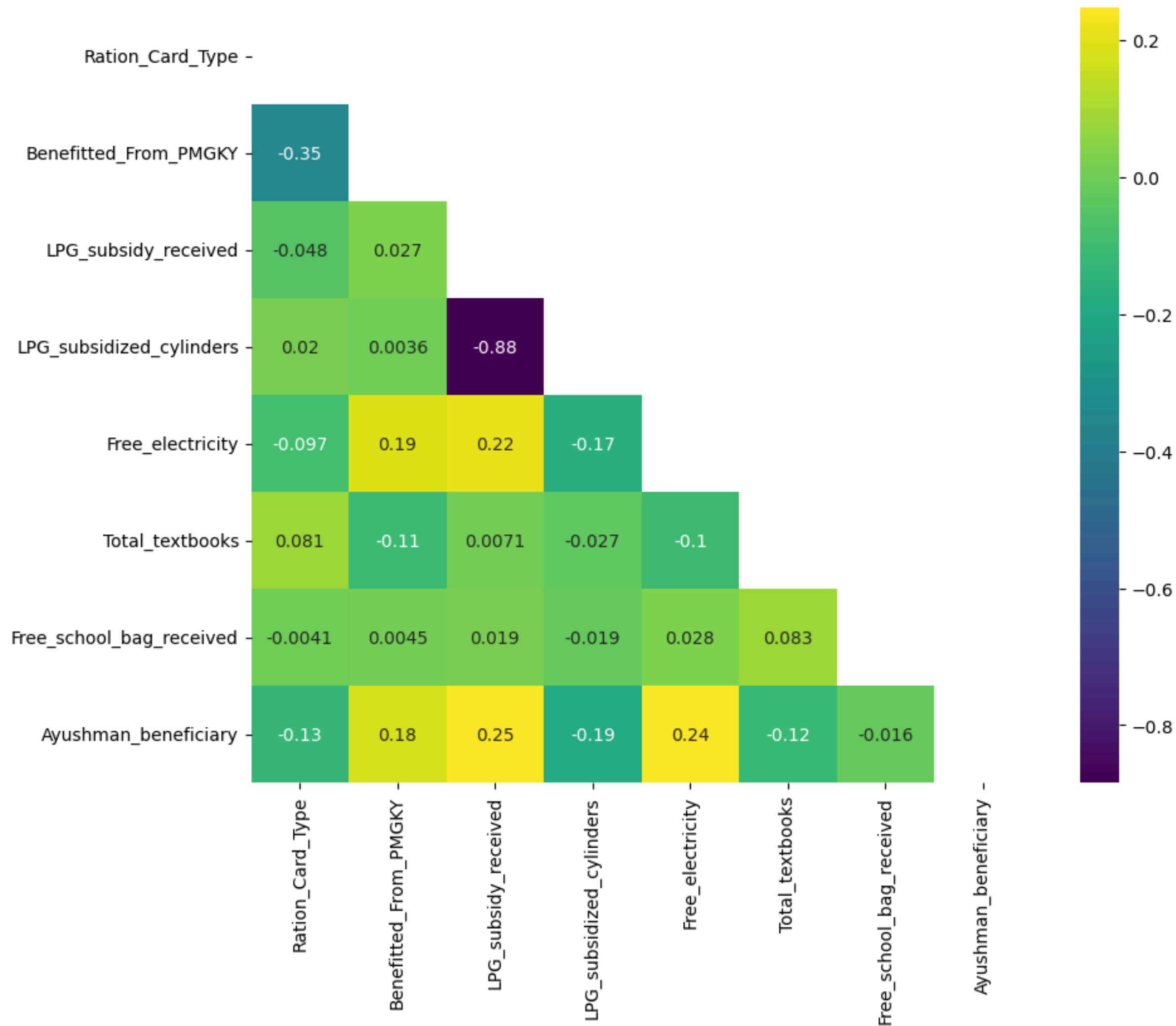
Online Activity Correlation Matrix



Professional Field Correlation Matrix Heatmap



Direct & Indirect Welfare Correlation Matrix



Thank You

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<https://github.com/Rudra-G-23/rural-financial-inclusion-govt-scheme-recommendation/tree/main/notebooks/correlations-all-levels>